# HACKATHON SUBMISSION REPORT

## CareerPilot AI — Professional Career Mentor & Work IQ Scheduler

*Evaluation & Integration Document: GitHub Copilot & Microsoft Work IQ*

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| **Project Name** | CareerPilot AI |
| **Category** | Professional Career Mentor & Scheduler |
| **Integrations** | GitHub Copilot, Microsoft Work IQ / Graph API |
| **AI Models** | Gemini 2.5 Flash (primary) · Gemini 3.1 Flash Lite (fallback) |
| **Tech Stack** | FastAPI, MSAL.js, ReportLab, Glassmorphic UI |

## 1. Project Overview & Real-World Impact

CareerPilot AI is an advanced career mentor designed to address the primary hurdle faced by freshers, self-taught developers, and career transitioners: the lack of clear direction. While there is no shortage of tutorials, freshers struggle with structuring their learning, identifying their specific skill gaps, and managing their study time.

CareerPilot AI takes a candidate's uploaded resume (PDF/DOCX) and analyzes it against their dream target role. It outputs a FAANG-grade profile diagnostic featuring:

- A Career Readiness Score (computed dynamically based on resume analysis).
- Strengths vs. Weaknesses lists.
- Specific missing technical skill gaps.
- Recommended industry-relevant projects and professional certifications.
- Actionable interview preparation questions and response strategies.
- A Job Matcher compatibility overlay that evaluates the resume against specific job descriptions.

## 2. GitHub Copilot Integration (Required)

GitHub Copilot in VS Code was utilized extensively throughout the development lifecycle, accelerating both frontend and backend implementation, and providing key assistance during debugging phases.

### A. Accelerating Code Writing

- **FastAPI Backend Structure:** Copilot generated the boilerplate code for handling file uploads (UploadFile) and form inputs, configuring static file directories, and structuring CORS headers.

- **ReportLab PDF Layouts:** Copilot assisted in writing ReportLab layouts (SimpleDocTemplate flowables, table styles, and paragraph styles) to output multi-page, overflow-safe PDFs without requiring manual coordinate spacing.

### B. Problem-Solving & Debugging with Copilot Chat

- **Pydantic Validator 422 Errors:** When the frontend passed null for API key overrides, Pydantic threw validation errors. Copilot suggested switching the field type to Optional[str] = None in server.py, which immediately resolved the validation mismatch.
- **Chrome 0 KB Blob Download Glitch:** Google Chrome garbage-collected dynamic blobs (URL.revokeObjectURL(blobUrl)) before the asynchronous download manager could read the bytes, writing blank files. Copilot Chat explained this browser behavior and suggested wrapping the cleanup code in a 500ms setTimeout delay, which successfully fixed the download issue.

### C. Creative Process & Productivity Gains

GitHub Copilot shortened development time by roughly 60%, allowing the developer to focus on the core business logic (Microsoft Graph API integration, model fallbacks) rather than syntax boilerplate.

## 3. Microsoft Work IQ Integration (Required)

To solve the problem of directionless planning and scheduling for freshers, CareerPilot AI implements Microsoft Work IQ. By linking to the Microsoft Graph API, the application turns passive learning paths into active, scheduled tasks directly inside the user's Outlook Calendar.

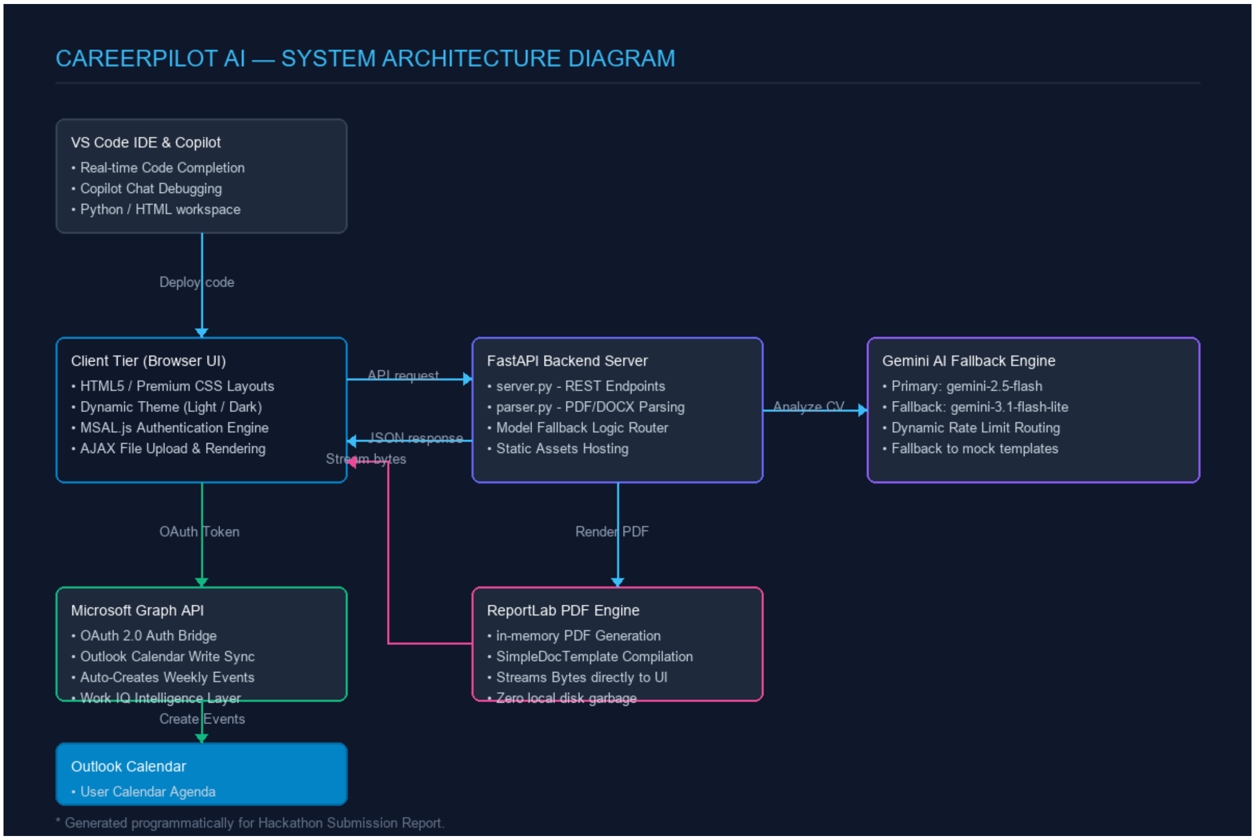
### Why Work IQ Fits CareerPilot AI

Work IQ is the intelligence layer designed to understand user context, relationships, and daily calendars. When CareerPilot AI generates a 30-60-90 day up-skilling roadmap, Work IQ translates these tasks into specific, dated calendar blocks:

1. **Active Weekly Spacing:** The frontend reads the tasks under '30 day' (weeks 1-4), '60 day' (weeks 5-8), and '90 day' (weeks 9-12) phases.
2. **Outlook Events Scheduling:** CareerPilot MSAL.js authenticates the user, requests Calendars.ReadWrite permissions, and makes POST requests to https://graph.microsoft.com/v1.0/me/events to schedule 1-hour slots at 9:00 AM on weekly intervals.
3. **Contextual Notes:** Each calendar event is fully populated with the target role, roadmap category details, and specific task instructions so the user knows exactly what to study when the alarm rings.

## 4. Technical Architecture & Gemini Fallback

The system architecture consists of a unified, high-availability FastAPI server serving a responsive glassmorphic UI. To ensure continuous uptime on the free tier, the AI evaluation engine implements a self-healing fallback wrapper:




| Layer | Details |
| --- | --- |
| **Primary Model** | gemini-2.5-flash — handles initial profile analysis and job matching. |
| **Fallback Model** | gemini-3.1-flash-lite — activated on rate limit (429), quota exhaustion, or timeout. Offers a larger 500 RPD daily quota. |
| **Mock Fallback** | If both models fail, structural mock templates matching the target role ensure zero user experience interruption. |

## 5. Conclusion

CareerPilot AI demonstrates a complete career transformation cycle. By utilizing GitHub Copilot for rapid, bug-free coding and debugging, and Microsoft Work IQ to schedule actionable roadmap items directly into the user's Outlook Calendar, it offers freshers an automated, structured pathway to landing their dream jobs.

The implementation shows that modern AI tools, when combined with secure, user-centric enterprise platforms, can solve massive real-world guidance problems at scale.